

Emanuele Rossi

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Researcher working on machine learning for animal communication, continuing a broader interest in how ML recovers structure in complex systems: previously social graphs (Temporal Graph Networks, SIGN) at Twitter and Imperial College London, and generative models for biomolecules at VantAI.

Education

Imperial College London Oct 2020 – Feb 2024

Ph.D. in Computer Science – Supervisor: Prof. Michael Bronstein

- Research focused on **graph machine learning methods** for large real-world systems

University of Cambridge Oct 2018 – Jun 2019

Master in Advanced Computer Science, Distinction (4.0 GPA)

Imperial College London Oct 2015 – Jun 2018

Bachelor in Computer Science, First Class Honors (4.0 GPA)

Experience

Postdoctoral Researcher Rome (remote)

GLADIA Research Group [🔗](#), Sapienza University of Rome

Oct 2025 – Present

- Developing machine learning methods to decode animal communication. Supervisor: Prof. Emanuele Rodolà

Machine Learning Researcher

VantAI

NYC (remote)

Jan 2024 – Aug 2025

- **Co-led development of Neo-1 [🔗](#)**, a state-of-the-art all-atom latent diffusion model for multimodal structure prediction and de-novo generation of biomolecular complexes, trained on 100+ GPUs.

Machine Learning Researcher

Twitter

London

Jun 2019 – Feb 2023

- Developed **graph ML models** for Twitter's social graph and deployed a recommendation model that generated **1.2M additional daily follows**

Machine Learning Researcher

Fabula AI

London

Mar 2019 – Jun 2019

- Developed graph ML models for fake news detection as part of Fabula AI's small research team before **Twitter acquired the company [🔗](#)**

Software Engineering Intern

Google

California

Jul 2017 – Oct 2017

Selected Publications

[Full list](#)

- *E. Rossi et al.* Temporal Graph Networks for Deep Learning on Dynamic Graphs [🔗](#). *ICML GRL Workshop 2020*. **1,700+ citations** and **1,000+ GitHub stars [🔗](#)**.
- *E. Rossi*, F. Frasca* et al.* SIGN: Scalable Inception Graph Neural Networks [🔗](#). *ICML GRL Workshop 2020*. **600+ citations** and used in **production at Airbnb [🔗](#)**.
- *B. Chamberlain, S. Shirobokov, E. Rossi et al.* Graph Neural Networks for Link Prediction with Subgraph Sketching [🔗](#). *ICLR 2023, Oral (top 5%)*. **200+ citations**.
- *E. Rossi et al.* Edge Directionality Improves Learning on Heterophilic Graphs [🔗](#). *Learning on Graphs 2023*.

Leadership

Co-Founder, LeadTheFuture ([Forbes article \[🔗\]\(#\)](#))

Jun 2018 – Mar 2023

- Built a non-profit mentoring community with more than **200 mentors** and **500 selected mentees**